

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I G. Poojitha (192425337) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Poojitha

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (59–50%)
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.

SIMATS ENGINEERING

Assessment Tool 2 — AI Application Research Report

AI-Based Autonomous Delivery Robot

Domain: *Robotics / Logistics*

1. Title of AI Application

AI-Based Autonomous Delivery Robot — a self-driving robot that carries small parcels, food, and groceries from a store or hub directly to the customer's doorstep without a human driver.

2. Domain Selection

Domain: Robotics, with strong overlap into Logistics and Retail. The system also touches Autonomous Vehicles technology, since it uses the same core navigation principles as self-driving cars, applied at a smaller, sidewalk scale.

3. Introduction and Background

An autonomous delivery robot is a small wheeled or legged machine that moves on sidewalks or roads to deliver items over short distances. It uses cameras, sensors, and AI software to see its surroundings, plan a route, and avoid people, vehicles, and obstacles. This technology started as a research idea in universities studying self-driving navigation, and has now grown into a real commercial service used by food delivery companies, retail stores, and university campuses. The purpose of the robot is to make last-mile delivery — the final short trip from a hub to the customer — faster, cheaper, and available even when human delivery staff are limited.

4. Problem Identification

Traditional last-mile delivery depends on human couriers, which becomes expensive and slow as delivery demand grows. Businesses face rising delivery costs, a shortage of delivery workers, traffic delays, and inconsistent delivery times, especially in dense cities and university campuses. Customers expect fast, low-cost, and contactless delivery, which is difficult for companies to guarantee using only human drivers.

5. Need for AI Solution

A simple pre-programmed machine cannot handle delivery on its own, because streets and sidewalks are unpredictable — pedestrians move randomly, obstacles appear suddenly, and routes can be blocked. AI is needed so the robot can sense its environment in real time, recognize objects such as people, vehicles, and curbs, and make its own decisions about the safest and shortest path. AI also allows the robot to learn from past deliveries and improve its route choices over time, something a traditional rule-based machine cannot do.

6. AI Technologies Used

- Computer Vision — to detect pedestrians, vehicles, obstacles, and traffic signals using onboard cameras.
- Machine Learning — to improve route choices and obstacle prediction based on past delivery data.
- SLAM (Simultaneous Localization and Mapping) — to build a live map of the surroundings while tracking the robot's own position.

- Path Planning / Search Algorithms (such as A*) — to calculate the shortest and safest path to the destination.
- Sensor Fusion — combining camera, LiDAR, ultrasonic, and GPS data for reliable perception.

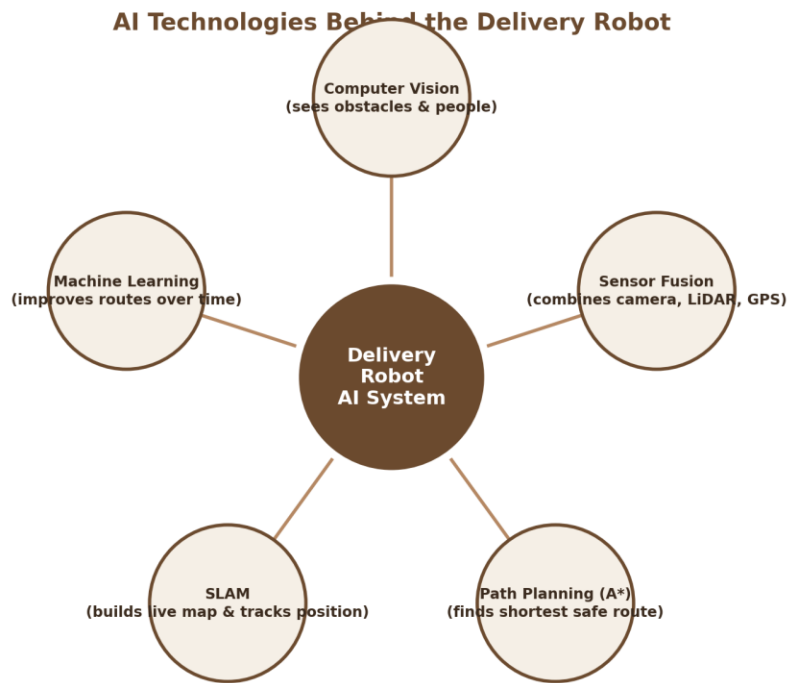


Figure 1: The five AI technologies working together inside the delivery robot.

7. Working Mechanism

Working Mechanism — How the Robot Delivers

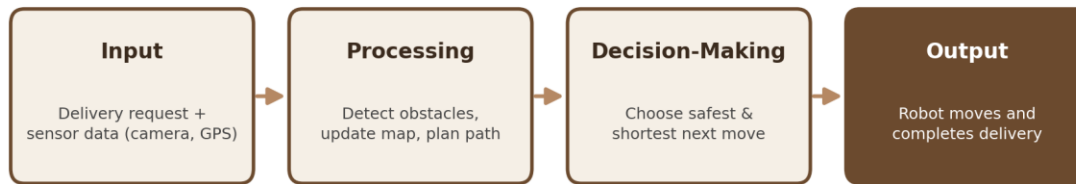


Figure 2: The four-step flow the robot follows for every delivery.

Input: The robot receives a delivery request with a pickup point and a drop-off address, along with live sensor data (camera images, GPS position, distance sensors).

Processing: The onboard AI system detects obstacles and pedestrians using computer vision, updates its map using SLAM, and calculates the best path using a search algorithm. It continuously re-checks this path as new obstacles appear.

Decision-making: Based on the path cost and safety checks, the robot decides whether to move forward, stop, turn, or wait for a pedestrian or vehicle to pass.

Output: The robot physically moves along the chosen path, crosses roads at safe points, and finally reaches the destination to complete the delivery.

8. Data Sources and Processing

The robot collects data from onboard cameras (video/images), LiDAR and ultrasonic sensors (distance readings), GPS (location), and wheel encoders (speed and movement). This raw data is processed in real time using computer vision models to identify objects, and using mapping algorithms to build a picture of the surrounding

area. Historical delivery data — such as which routes were fastest or had frequent obstacles — is also stored and used to improve future route planning.

9. Real-World Implementation

Autonomous delivery robots are already used commercially in several countries. Starship Technologies operates a large fleet of sidewalk robots across many cities and university campuses worldwide, completing millions of deliveries for groceries and food orders. Serve Robotics runs similar robots for restaurant deliveries in major U.S. cities, and food delivery platforms such as Uber Eats have begun deploying sidewalk delivery robots in cities abroad. These deployments show that the technology has moved from research prototypes to everyday commercial use, particularly for short-distance food, grocery, and parcel delivery.

10. Impact and Benefits

- Lower delivery cost per order compared to human couriers, especially for short distances.
- Faster delivery for nearby locations, since robots can operate continuously without breaks.
- Reduced traffic congestion and emissions compared to car-based delivery.
- Contactless delivery, which many customers prefer for hygiene and convenience.
- Ability to operate in labor shortages, keeping delivery services running when human staff are limited.

11. Challenges and Limitations

- Technical: Robots can struggle with heavy rain, snow, poor lighting, or very crowded sidewalks.
- Safety: Ensuring the robot never collides with pedestrians, cyclists, or vehicles is a continuous engineering challenge.
- Privacy: Onboard cameras recording public streets raise questions about surveillance and data storage.
- Security: Robots can be targets for theft, tampering, or vandalism since they operate unattended in public spaces.
- Regulatory: Many cities are still creating laws and sidewalk-use rules for delivery robots, which can slow expansion.

12. Future Scope and Emerging Trends

- Expansion beyond food and groceries into pharmacy, retail, and internal corporate logistics deliveries.
- Better all-weather navigation using improved sensor fusion and AI models.
- Multi-robot coordination, where fleets of robots share map and traffic data to avoid each other efficiently.
- Integration with drones for a hybrid ground-and-air last-mile delivery network.
- Use of more advanced AI (such as reinforcement learning) to let robots learn safer behavior directly from experience.

13. Conclusion

The AI-based autonomous delivery robot is a practical example of how AI, computer vision, and search algorithms can solve a real, everyday logistics problem. By

combining sensing, mapping, and intelligent path planning, these robots can deliver items faster, more cheaply, and more reliably than traditional methods in many situations. While challenges around safety, weather, and regulation remain, real-world deployments by companies such as Starship Technologies and Serve Robotics show that this AI application has already moved from research into large-scale practical use, and is expected to keep expanding across cities and industries.

14. References

1. Company websites and public deployment reports of autonomous delivery robot providers (e.g., Starship Technologies, Serve Robotics).
2. Industry and technology news articles on autonomous delivery robots and last-mile logistics (2026).
3. General robotics literature on SLAM, computer vision, and path-planning algorithms such as A*.
4. Class notes and reference material for Artificial Intelligence and Expert Systems.